Diagnosing Failures of Neural Quantum States

Controlled interventions on representation, optimization, sampling, and quantum geometry

Don van Beusichem

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Abstract

A stalled neural quantum-state calculation does not identify its own failure mechanism. We construct a controlled diagnostic study using exact spin-chain references, manually differentiated neural ansätze, supervised state fitting, variational optimization, frozen-distribution Monte Carlo, and quantum-geometric-tensor interventions. The study retains 129 trained models, 80 sampling repetitions, and 800 independent sampling batches for one-step geometry tests. A positive-amplitude ansatz has a proven infidelity floor of one half for the selected frustrated-chain target in the stated basis, whereas a signed network reaches the ground state to numerical precision at sufficient width. Lower-energy parameter witnesses establish optimization shortfalls for smaller signed networks without equating a supervised fit with a representation optimum. In a nearly degenerate Ising chain, exact-sum energy training gives infidelity near one half despite an energy error below $10^{-6}J$; symmetry projection restores high state fidelity. Frozen local Metropolis chains report essentially exact energy and zero local-energy variance while missing one magnetization sector. Finite-sample geometric updates can acquire uphill directions, while insufficient regularization also causes deterministic finite-step overshoot. Finally, parameter rescaling changes a thresholded QGT rank from 80 to 24 for an unchanged state family. These results support a diagnostic protocol based on constructive controls and observable checks, rather than a single energy, variance, or rank statistic. No general automatic classifier of failure mechanisms or scalable optimization advantage is claimed.

1. Introduction

Neural quantum states encode many-body wavefunctions with a parametrized function of spin or occupation configurations. Their numerical accuracy depends on at least three different ingredients: a suitable variational family, an optimizer that can exploit that family, and reliable sampling of the current state. A plateau in the training energy is compatible with failure in any of them. Increasing width or changing the learning rate without identifying the cause can therefore spend computational effort on the wrong intervention.

There is a fourth issue: the optimized objective may weakly constrain the physical quantity used to judge the result. In a finite system with a very small gap, two states with markedly different global fidelity can have almost indistinguishable energy. In that situation, low energy is a valid achievement for an energy objective, but it does not certify recovery of a particular reference vector or symmetry sector.

This study develops a sequence of small, reproducible interventions that make these distinctions explicit using configuration-conditioned spin wavefunctions. Hamiltonians, exact references, neural derivatives, samplers, and stochastic-reconfiguration updates are implemented directly with NumPy and SciPy. Exact state-space enumeration is deliberately retained so that approximation, optimization, and sampling can be changed independently.

The central contribution is an experimental protocol and a set of controlled examples, not a claim that these failure mechanisms or the QGT are newly discovered. We use a provable sign restriction for a genuine representation obstruction; compare actually obtained parameter vectors to certify some optimization shortfalls; hold a state fixed to isolate sampler behavior; and independently perturb the estimated metric and gradient to examine finite-sample update errors. A parameter-coordinate stress test then checks which geometric diagnostics depend on a numerical convention.

The results favor a cautious interpretation of familiar diagnostics. High QGT rank is useful evidence about local tangent directions, but is not a certificate of global representability. Low energy and small local-energy variance are informative about a state, but can be insensitive to a sampler missing an entire mode. A positive-definite metric ensures a descent direction only when the true gradient is used, and a descent direction does not guarantee that a finite nonlinear step lowers the energy. Each statement has a concrete counterfactual test below.

2. Background and related work

Neural representations of quantum states were established as a variational many-body method by Carleo and Troyer [1]. Subsequent work has explored the effects of sign structure, sampling, architecture, and optimization. Park and Kastoryano systematically examined RBM limitations for non-stoquastic spin Hamiltonians, including both representation and sampling difficulties [2]. The present positive-amplitude restriction is a particularly simple, analytically inspectable control within that broader setting; it is not a statement about complex RBMs in general.

Dash and collaborators, including Vicentini, investigated the practical efficiency of neural states using supervised infidelity minimization and QGT spectra [3]. That work motivates exact-sum supervision as a diagnostic and explicitly treats rank as a local probe. Our parameter-rescaling and sign-restriction examples clarify the limits of using a raw rank statistic outside a controlled parametrization. They do not refute the usefulness of rank within a specified model family and numerical convention.

Optimization developments include minimum-step stochastic reconfiguration [4] and block approximations to the QGT [5]. A recent preprint interprets finite-sample SR as statistically regularized tangent-space regression and studies shift-dependent overfitting [6]. We therefore do not claim that the shift is merely a numerical stabilizer, or that a spectral filter is intrinsically new. The comparisons below use small explicit metric matrices and simple interventions rather than reproduce those large-scale algorithms.

Known symmetries can be built into neural states to select sectors [7]. Sampling methods also remain an active subject; neural importance resampling is one example of a method designed to improve sampling without imposing an autoregressive form on the target state [8]. Our global-flip proposal is much simpler and exploits an exact spin-flip symmetry. It serves as a controlled mixing repair, not a general-purpose replacement for advanced samplers.

3. Models, references, and implementation

3.1 Spin Hamiltonians

The transverse-field Ising Hamiltonian is

$$H_{\mathrm{I}} = -\sum_{i=0}^{N-1} \sigma_i^z \sigma_{i+1}^z - h \sum_{i=0}^{N-1} \sigma_i^x,$$

with periodic indices and coupling $J = 1$. The primary systems use $N = 8$ and $h = 0.2, 1, 2$, denoting ordered, critical-coupling, and paramagnetic examples. An additional $N = 10$, $h = 1$ calculation

checks one larger exact state space. These labels refer to the conventional infinite-chain coupling regimes; no finite-size phase transition is claimed.

The frustrated model is

$$H_{12} = \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1} + J_2 \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+2}, \qquad \mathbf{S}_i = \boldsymbol{\sigma}_i/2,$$

with $N = 8$ and $J_2 = 0.7$. It is restricted to $S_z = 0$, with 70 configurations. A Marshall sign transformation multiplies each basis vector by $(-1)^{n_{\downarrow,A}}$, where A is the even sublattice. All sign-restriction statements specify this basis. A separate coupling scan checks how the positivity bound changes with basis and J_2; it is not a phase-diagram calculation.

Ising calculations enumerate all 2^N configurations. Spin configurations are stored as bit strings and converted to $\sigma_i = \pm 1$. Sparse Hamiltonians are assembled by explicit spin flips or exchanges. Dense diagonalization provides reference eigenvalues and normalized target vectors. For the Ising ground state, the positive even-parity vector is selected consistently; no symmetry is imposed on the main unprojected RBM fits.

3.2 Independent reference checks

An independent tensor-product construction verifies all matrix elements for small Ising and Heisenberg examples. Ising ground energies are also checked against the finite-chain even-parity free-fermion expression

$$E_0 = -\sum_{m=0}^{N-1} \sqrt{1 + h^2 - 2h\cos k_m}, \qquad k_m = (2m+1)\pi/N.$$

The checked Hamiltonian matrices agree exactly at printed precision, and the analytic ground-energy differences are of order $10^{-14}J$. The checks distinguish matrix assembly from neural optimization. They do not validate thermodynamic-limit extrapolation or large-system scaling.

3.3 Positive RBM and signed neural ansatz

The real positive RBM is

$$\psi_\theta(\sigma) = \exp(a \cdot \sigma) \prod_{j=1}^{m} 2\cosh\left(b_j + \sum_i W_{ij}\sigma_i\right).$$

It has $N + m + Nm$ parameters and is strictly positive at finite real parameters. Log amplitudes are evaluated stably before normalization. Widths $m = 4, 8, 16$ are tested for the eight-site systems; the ten-site comparison uses $m = 8$.

The signed model used for the frustrated chain is

$$v_\theta(\sigma) = d + \sum_{j=1}^{m} u_j \tanh\left(b_j + \sum_i W_{ij}\sigma_i\right), \qquad c_\theta = v_\theta/\|v_\theta\|.$$

Its real coefficients can change sign. It has $Nm + 2m + 1$ parameters and is evaluated only on the stated $S_z = 0$ sector. Width changes therefore alter both parameter count and optimization landscape. This is not an architecture-matched comparison to the RBM, and the purpose is not to identify a best general ansatz.

All normalized coefficients and their parameter Jacobians are calculated analytically. For the RBM, with logarithmic derivatives $O_a = \partial_a \log \psi$, the normalized-state Jacobian is

$$B_{\sigma a} = \partial_a c_\sigma = c_\sigma \left(O_a(\sigma) - \langle O_a \rangle_p \right), \qquad p_\sigma = c_\sigma^2.$$

For the signed network, the unnormalized amplitude Jacobian is projected orthogonally to c and divided by the amplitude norm. Finite differences validate both families. Sampling experiments and sampled logarithmic derivatives use positive states only, avoiding nodal divisions in the signed model.

3.4 Exact objectives and training budgets

The variational objective is $E = c^T H c$, with exact gradient

$$g_E = 2B^T(H - E)c.$$

The supervised diagnostic objective is infidelity to the exact target c_0,

$$\mathcal{I} = 1 - (c^T c_0)^2, \qquad g_{\mathcal{I}} = -2(c^T c_0)B^T c_0.$$

Each family/width/objective combination uses initialization seeds 0, 1, and 2. L-BFGS-B allows 1,200 iterations with explicit termination records. There are 96 main fits. Supervised fitting uses exact target labels and is not a scalable ground-state method. Crucially, its achieved loss is only an upper bound on the best achievable representation error: supervised optimization can fail too.

Nine additional energy fits use the even spin-flip projection

$$c_+(\sigma) \propto \psi_\theta(\sigma) + \psi_\theta(-\sigma).$$

This is a changed, symmetry-constrained family requiring two base evaluations, not a cost-free change to an otherwise identical RBM. Another 24 exact-sum runs compare manually implemented Adam and three SR variants over 500 update steps, giving 129 retained trained models in total. No end-to-end training run driven by local MCMC is claimed; sampling and stochastic updates are isolated in the separate controls below.

4. Diagnostic framework

4.1 A provable representation restriction

For a normalized real target u, define the positive and negative sign masses

$$P_+ = \sum_{u_\sigma > 0} u_\sigma^2, \qquad P_- = \sum_{u_\sigma < 0} u_\sigma^2.$$

For any normalized nonnegative vector c, Cauchy–Schwarz gives

$$|u^T c|^2 \le \max(P_+, P_-), \qquad \mathcal{I} \ge \min(P_+, P_-).$$

The supremum is approached by matching the larger sign component and suppressing the other. A finite positive RBM need not attain that supremum. In the chosen frustrated-chain target, both masses are one half to numerical precision, giving an infidelity lower bound of one half for every real positive RBM, irrespective of width or optimizer.

This is a restriction of an ansatz in a basis, not a basis-independent impossibility theorem for the physical state. A known sign transform can alter the masses; a general sign transform based on the unknown exact target would be an oracle rather than a practical solution.

4.2 Constructive optimization witnesses

Let θ_e be an energy-trained state and θ_w any obtained parameter vector in the same family and width. If $E(\theta_w) < E(\theta_e)$, then

$$E(\theta_e) - \inf_{\theta} E(\theta) \geq E(\theta_e) - E(\theta_w) > 0.$$

The right-hand side is a certified lower bound on optimization shortfall, even when the true variational minimum is unknown. We use supervised-fit parameters as matched witnesses and also retain a pooled best-known witness across seeds and objectives. This establishes some optimizer shortfalls without assuming that the supervised objective found the best representable state.

The absence of a better witness proves nothing. Similarly, a supervised state can have lower infidelity but higher energy because it trades excited-state components differently. Such a comparison is an objective-sensitivity observation, not a lower-energy witness.

4.3 Objective sensitivity near degeneracy

For a unique exact ground state and gap $\Delta = E_1 - E_0$,

$$\mathcal{I} \leq \min\left(1, \frac{E - E_0}{\Delta}\right).$$

The bound becomes weak when Δ is small. The eight-site ordered Ising example has a finite-size gap about $1.05 \times 10^{-6} J$. Hence a total energy error below $10^{-6} J$ need not constrain global fidelity tightly. We record magnetization and spin-flip expectation as well as energy and infidelity, distinguishing a useful low-energy state from recovery of the specified finite-system eigenvector.

4.4 What the QGT measures locally

For these real normalized states, the QGT metric is

$$S = B^T B = \mathrm{Cov}_p(O, O).$$

It is positive semidefinite, and its image describes local normalized-state tangent directions. We also decompose the residual $r = (H - E)c$ into its projection onto the column space of B and the orthogonal remainder. Both the absolute remainder and its fraction of $\|r\|$ are recorded at reference coordinates and a stated singular-value tolerance. A small denominator can make a relative remainder misleading near an eigenstate.

Neither tangent rank nor a local residual projection alone determines global representability. For example, the positive log-table family $c_i \propto e^{\theta_i}$ has

$$S = \mathrm{diag}(p) - pp^T,$$

which has rank $D - 1$ whenever every $p_i > 0$. It spans the full local tangent dimension inside the positive orthant, yet it cannot approach a sign-changing target beyond the bound above. In the 70-dimensional sector, this gives a full formal rank of 69 alongside an infidelity floor of one half. This elementary counterexample clarifies the difference between local dimension and a global sign constraint.

4.5 Frozen sampling isolates mixing

We freeze the exact positive even ground state of a ten-site Ising chain at $h = 0.2$. Sampling its known Born distribution removes ansatz and optimization errors. Four controls use exact independent draws, local single-spin Metropolis updates initialized up, the same local updates initialized down, and Metropolis updates with a 5% whole-chain spin-flip proposal probability.

Each repetition discards 1,000 sweeps and retains 4,096 samples; one sweep contains N attempted proposals. There are 20 seeds per method. Up/down cases are symmetry-paired with common random numbers and are not independent studies. The global proposal is accepted with probability one for the exact symmetric target and preserves its stationary distribution.

We compare magnetization, empirical total variation and Jensen–Shannon divergence, sign changes of magnetization, energy estimates, and local-energy variance. The displayed naive standard error uses the independent-sample formula and is deliberately not a valid autocorrelation-corrected uncertainty estimate. It illustrates how a seemingly precise within-mode estimate can miss a global mixing failure.

4.6 Independent interventions on metric and gradient errors

At steps 5 and 25 of one exact critical-Ising RBM fit, we freeze the parameters and construct the population metric S, gradient g, and Born distribution. For $M = 16, 64, 256, 1024, 100$ independently seeded batches are drawn from that exact distribution. Thus this experiment contains finite-sample covariance error but no Markov-chain mixing error.

Sample matrices and gradients use centered plug-in moments with normalization $1/M$:

$$\widehat{S} = \frac{1}{M}\sum_k (O_k - \bar{O})(O_k - \bar{O})^T, \qquad \widehat{g} = \frac{2}{M}\sum_k (O_k - \bar{O})(E_{\mathrm{loc},k} - \bar{E}_{\mathrm{loc}}).$$

These plug-in covariances include the usual finite-sample normalization bias. Their rank cannot exceed $M - 1$, the parameter count, or the number of distinct configurations minus one. We record all three relevant counts.

Matched controls replace only S, only g, or both. The reference update is $d = -(S + \lambda I)^{-1} g$, with $\lambda = 10^{-3}\bar{s}$ and $\bar{s} = \mathrm{Tr}(S)/P$. Additional matched population and sampled updates use shifts $10^{-6}\bar{s}$ and $10^{-2}\bar{s}$. The scale $\bar{s}$ is the exact population scale in this diagnostic, deliberately avoiding another scale-estimation confound.

We also examine a diagonal shift, a block approximation partitioned into visible biases, hidden biases, and weights, and an exploratory split-sample filter. The block is an RBM parameter-group approximation, not a reproduction of a deep-network layer-wise method [5]. Diagonal shifts include a tiny isotropic floor for numerical stability.

Every candidate direction is rescaled to the same *true linearized* displacement $\|B\delta\theta\| = \rho$, for $\rho = 0.002, 0.01, 0.02$. This is an oracle normalization used for controlled comparisons, not a deployable sampled trust-region algorithm. We record the exact first-order energy change, exact finite energy change, direction cosine against the population reference step, and actual phase-invariant chord distance between states.

A positive-definite noisy metric combined with the true gradient always gives $g^T d \leq 0$. Once the gradient is replaced by an estimate, that guarantee is absent. Separately, a first-order descent direction can increase energy after a finite step because the neural state is nonlinear. Matched deterministic small-shift controls distinguish that effect from sampling noise.

4.7 Coordinate-rescaling stress test

For an invertible diagonal parameter change $\theta = D\phi$,

$$S_\phi = D^T S D, \qquad g_\phi = D^T g, \qquad B_\phi = BD.$$

The represented states and exact tangent dimension are unchanged. However, thresholding eigenvalues relative to the largest one can produce a different numerical rank. Our rank rule counts $\lambda_i > \max(10^{-8}\lambda_{\max}, 10^{-12})$. A column-normalized metric $C_{ab} = S_{ab}/\sqrt{S_{aa}S_{bb}}$ is invariant under positive diagonal rescaling of active columns, but not under arbitrary nonlinear reparametrization.

We choose fixed random exponents and scale extents 0, 1, 3, and 6 decades. Isotropic trace-scaled ridge updates are compared with diagonal-shift updates. The latter transform covariantly under positive diagonal rescaling in the ideal arithmetic limit. Linear systems are solved in column-normalized coordinates to avoid confusing that identity with unstable raw matrix inversion.

5. Results

5.1 Exact references and implementation checks

Independent tensor-product Hamiltonians agree elementwise with the bit-string implementations in the checked systems. The largest finite-difference parameter-gradient discrepancy is approximately 1.2×10^{-9} for the unprojected models, and the normalized-state tangent is orthogonal to the state within approximately 3.6×10^{-16}. The QGT covariance identity agrees within 2.3×10^{-16}. A separate projected-RBM derivative check agrees within 3.3×10^{-10}.

These checks establish numerical consistency of the chosen finite models. They do not imply optimizer convergence. Of the 96 main L-BFGS-B fits, 39 terminate successfully under their numerical criterion; the remainder reach the iteration cap. Both outcomes are retained. Some converged fits are poor, while some cap-limited fits are accurate, emphasizing that termination status and physical error are distinct quantities.

Table 1. Frozen critical-Ising checkpoints for the QGT study. Residual norms use $J = 1$ units. The tangent remainder is computed with the stated threshold in the reference coordinates.

Step	Energy error	Infidelity	QGT rank	Residual norm	Orthogonal fraction
5	0.4505	0.4074	80	1.439	0.09408
25	0.01492	0.003732	80	0.3248	0.2874

From step 5 to step 25, both energy error and residual norm decrease. The fraction of residual outside the retained tangent space increases, while its absolute norm decreases from about 0.135 to 0.0933. Reporting only that fraction would therefore suggest a deterioration that the absolute residual does not support. This is a local, threshold-dependent diagnostic, not a proof about the best state anywhere in the ansatz.

5.2 Width, objective, and a low-gap failure of energy certification

Table 2. Exact-sum Ising fits at width eight. Entries are means with population standard deviations over three seeds. Energy errors are total-system errors in units of J; infidelity is dimensionless. All unprojected fits have the same 1,200-iteration cap.

System	Training objective	Energy error	Infidelity
N=8, h=0.2	energy	7.85e-07 (5.1e-08)	0.5 (3.5e-06)
N=8, h=0.2	infidelity	3.3e-06 (5.7e-07)	4.41e-07 (6.1e-08)
N=8, h=1	energy	9.02e-05 (1.4e-05)	8.66e-06 (1.4e-06)
N=8, h=1	infidelity	0.00013 (1.9e-05)	1.2e-05 (1.9e-06)
N=8, h=2	energy	1e-06 (5.6e-07)	6.48e-08 (3.9e-08)
N=8, h=2	infidelity	1.09e-06 (2.9e-07)	6.82e-08 (1.8e-08)
N=10, h=1	energy	0.000539 (1.4e-05)	6.82e-05 (2.6e-06)
N=10, h=1	infidelity	0.000527 (4.4e-05)	6.09e-05 (6e-06)

At the ordered coupling, energy-trained states have mean error about $7.85 \times 10^{-7} J$ and infidelity approximately 0.5. Infidelity-trained states have error about $3.30 \times 10^{-6} J$ but infidelity approximately 4.41×10^{-7}. The supervised state is much closer to the specified target while slightly worse in energy. That comparison alone is not proof that the energy optimizer failed to minimize its objective.

The small finite-system gap explains why these statements coexist. The energy-trained state can put substantial weight in a nearly degenerate symmetry sector while keeping higher-energy contamination small. It is a good low-energy approximation for an energy-only task, but a poor approximation to the selected finite-size even eigenvector. No claim is made that a symmetry-broken low-energy state is physically meaningless in the thermodynamic ordered phase.

Symmetry projection changes this result substantially. At width eight, energy optimization of the even-projected family gives mean energy error 2.27e-07J and mean infidelity 3.53e-08. Magnetization vanishes to numerical precision and parity is one. The projection supplies correct sector information and changes the family and evaluation cost. Its success is therefore an example of a useful physical inductive bias, not evidence that representation was unchanged.

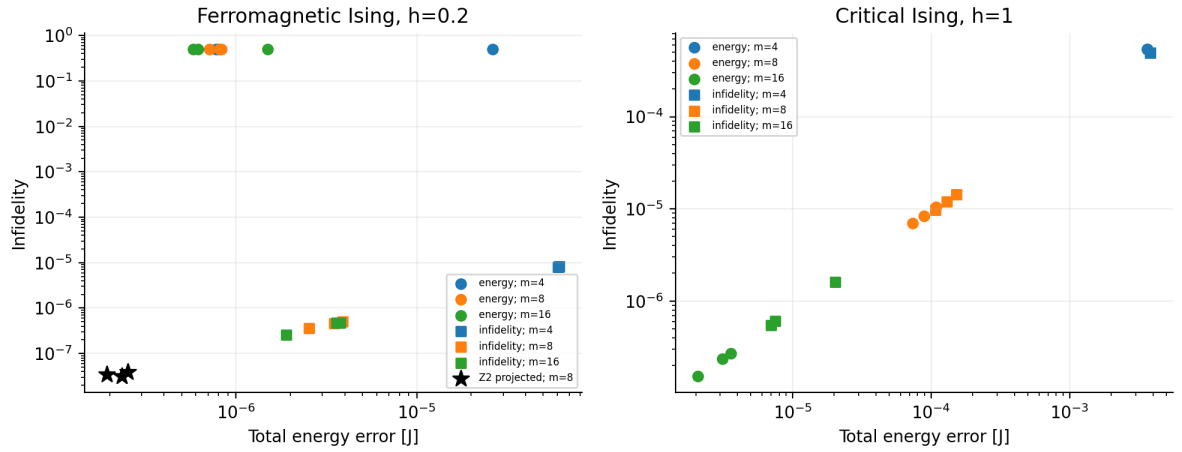


Figure 1: Energy error versus infidelity for the eight-site Ising fits. Colors identify width and markers distinguish the two objectives. The projected width-eight fits are marked separately in the ordered example. Three seeds are shown individually. Low energy alone does not identify a unique target vector near degeneracy.

5.3 A certified sign restriction and optimization witnesses

Table 3. Frustrated eight-site chain in the Marshall basis. Three-seed means and standard deviations are reported. The nonnegative-family infidelity bound of 0.5 applies to the RBM rows, not to the signed network.

Family	Width	Objective	Energy error	Infidelity
RBM	4	energy	0.29 (0.16)	0.582 (0.05)
RBM	4	infidelity	0.779 (0.22)	0.629 (0.074)
RBM	8	energy	0.405 (0.16)	0.617 (0.05)
RBM	8	infidelity	0.847 (0.46)	0.639 (0.14)
RBM	16	energy	0.405 (0.16)	0.617 (0.05)
RBM	16	infidelity	0.953 (0.6)	0.663 (0.18)
MLP	4	energy	0.491 (0.26)	0.718 (0.22)
MLP	4	infidelity	0.551 (0.052)	0.241 (0.017)
MLP	8	energy	0.0552 (0.035)	0.325 (0.25)
MLP	8	infidelity	0.00395 (0.0014)	0.00144 (0.00063)
MLP	16	energy	1.73e-13 (1.3e-13)	4.29e-13 (4.5e-13)
MLP	16	infidelity	0.000258 (0.00036)	0.000114 (0.00016)

No positive RBM fit violates the analytic bound. Increasing its width cannot remove that specific sign obstruction, although it can still change the quality and difficulty of optimization within the allowed family. Some runs remain far above the lower bound, which does not imply that the bound is attainable by a finite RBM of that width.

The signed width-sixteen network reaches the reference energy to approximately $10^{-13}J$ in the energy fits. Thus the physical target is accessible to a tested signed family. Width-eight signed energy fits are much less reliable, but supervised-fit parameters at the same width provide better energies. The matched lower bounds on their optimization shortfalls are seed 0: 0.008749J, seed 1: 0.05511J, seed 2: 0.08988J. These are constructive statements about attained parameters and do not require solving the global variational problem.

The reverse caution also matters. At width sixteen, energy fits outperform the retained supervised fits in both mean energy and mean infidelity. Supervised fitting is therefore a useful intervention, not an oracle that automatically reveals the representation floor. Reporting its loss as that floor would incorrectly attribute some optimization failures to the ansatz.

The sign restriction is basis dependent. In a separate $J_2 = 0$ calculation, the Marshall-basis target has no negative sign mass, whereas the original-basis positive-family lower bound is about 0.452. At $J_2 = 0.7$, both inspected bases have sign masses one half. These finite-system observations concern the specified basis and target; they are not a classification of all frustrated quantum states.

5.4 Sampler failure despite exact energy and zero variance

Table 4. Frozen exact Ising ground-state sampling. Each method has 20 repetitions with 4,096 retained configurations. The reference magnetization per spin is zero. JSD uses natural logarithms.

Sampler	Mean absolute magnetization error	Mean JSD	Runs with no sign change
iid	0.01114	0.0008939	0
local_up	0.9947	0.2164	20
local_down	0.9947	0.2164	20
global_flip	0.01672	0.001408	0

Every local-up and local-down repetition remains in its initial magnetization sector. Their mean magnetizations are approximately $+0.995$ and -0.995, respectively, even though the exact expectation

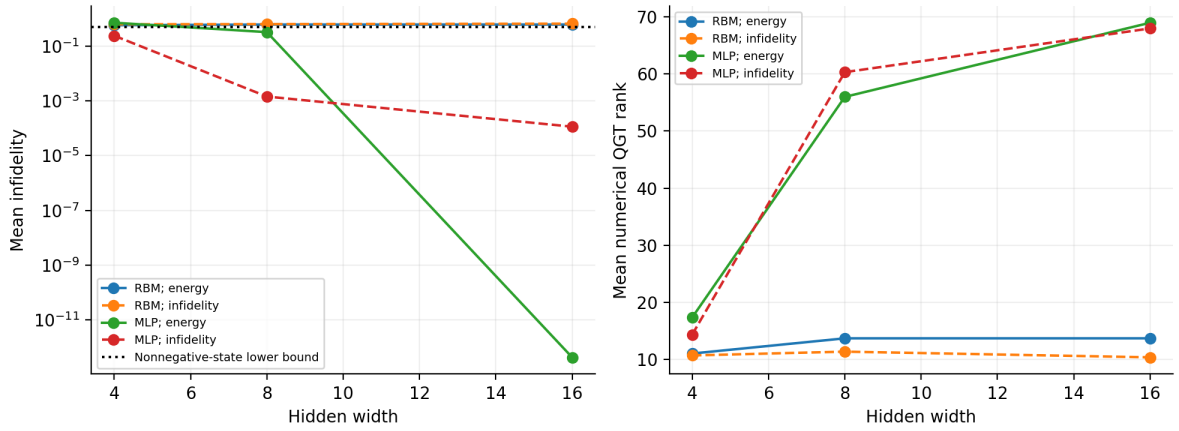


Figure 2: Frustrated-chain infidelity and thresholded QGT rank versus hidden width. Lines distinguish objective and family. The horizontal infidelity bound applies only to nonnegative states. The rank is local and varies with the achieved state; it is not a certified representation-capacity curve.

is zero. The empirical distribution misses roughly half the target probability mass. Independent draws and whole-chain proposals recover much smaller magnetization errors under the same retained-sample count.

Nevertheless, the mean local-energy estimate is correct to numerical precision for all methods, and observed local-energy variances are of order $10^{-30}J^2$. This follows directly from the exact eigenstate identity $E_{\mathrm{loc}}(\sigma) = E_0$ wherever the state has support. A biased sampler cannot be diagnosed from the fluctuations of a constant observable.

The naive within-chain standard error of magnetization is about 5.1×10^{-4} for the trapped chains, vastly smaller than their order-one error. That number is not an appropriate uncertainty estimate for the global mean. Its failure motivates checking mode-sensitive observables and distinct initializations rather than inferring mixing from energy stability.

A simple symmetry audit, $|\widehat{\langle M \rangle}| > 0.1$, flags all 20 trapped repetitions from each initialization and none of the independent/global-flip repetitions in this experiment. The threshold is an illustrative diagnostic, not a calibrated hypothesis test. When the *current trial state* is known to be exactly even, the zero odd-observable expectation follows without exact diagonalization. For an unprojected asymmetric trial state, imposing zero magnetization as a sampler truth would be incorrect.

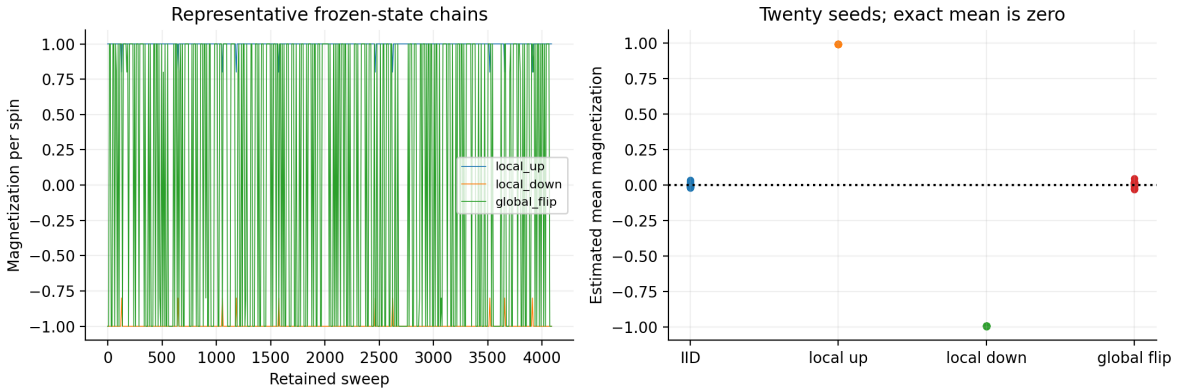


Figure 3: Representative retained magnetization traces and the estimated mean magnetization across twenty seeds. The local chains sample opposite modes depending on initialization. Whole-chain proposals restore mode changes. Every sampler here uses the same frozen exact state; none of these differences can be attributed to neural optimization.

5.5 Finite-sample rank is not representation rank

Table 5. Sampled metric ranks at checkpoint 25. The exact metric rank is 80 in the reference coordinates. The count of distinct configurations provides a tighter rank limit than sample count in some batches.

Samples	Distinct configurations	Metric rank	Relative metric error	Relative gradient error
16	10.5 (1.8)	9.5 (1.8)	0.6027	5.471
64	30.5 (3.5)	29.5 (3.5)	0.3052	2.871
256	76.7 (5.3)	74.9 (4.3)	0.1549	1.453
1024	147 (6.2)	80 (0)	0.07662	0.7361

A low sampled rank can arise solely because too few configurations were observed. In the experiment the underlying neural family, frozen parameters, and exact metric remain unchanged. Interpreting that sampled rank as a reduction in intrinsic representation capacity would therefore be a category error.

The local-tangent limitation is different again. The positive log-table example has full local rank 69 in the frustrated sector and still has the proven sign-induced infidelity floor. Together these examples show why a rank statistic needs a declared object: population or sampled metric, numerical or formal rank, fixed parametrization or rescaled coordinates, and local or global representability.

5.6 Noisy metric and gradient interact

Table 6. Matched metric/gradient interventions at checkpoint 25. The ridge is $10^{-3}\bar{s}$ and the true linearized displacement is 0.002. Entries are percentages of uphill directions or finite steps over 100 batches, and mean direction cosine to the population reference.

Samples	Intervention	Linear uphill (%)	Finite uphill (%)	Direction cosine
16	population	0	0	1
16	metric only noise	0	0	0.2923
16	gradient only noise	0	0	0.2576
16	joint ridge	28	33	0.0389
256	population	0	0	1
256	metric only noise	0	0	0.704
256	gradient only noise	0	0	0.7357
256	joint ridge	0	0	0.8049

With the true gradient, a regularized positive-definite sampled metric cannot produce a first-order uphill direction; the observed zero rate is consistent with that algebraic identity. Gradient replacement alone also produces no first-order uphill directions in these retained batches, although it changes directions and can affect finite steps.

Replacing both quantities can produce uphill directions at small sample count. The phenomenon is therefore not adequately described as the metric having become indefinite: the regularized metric remains positive definite. The estimated gradient and the geometry used to precondition it jointly determine alignment with the true energy gradient. Increasing samples reduces this particular failure in the tested checkpoints, without implying a universal sample threshold for other states.

Alternative ridge strengths, a diagonal shift, and a parameter-group block approximation are all retained in the raw records. They can improve one-step behavior relative to an excessively small shift, but no single approximation is established as best over systems or training stages. The present scale is too small to evaluate a computational scalability advantage.

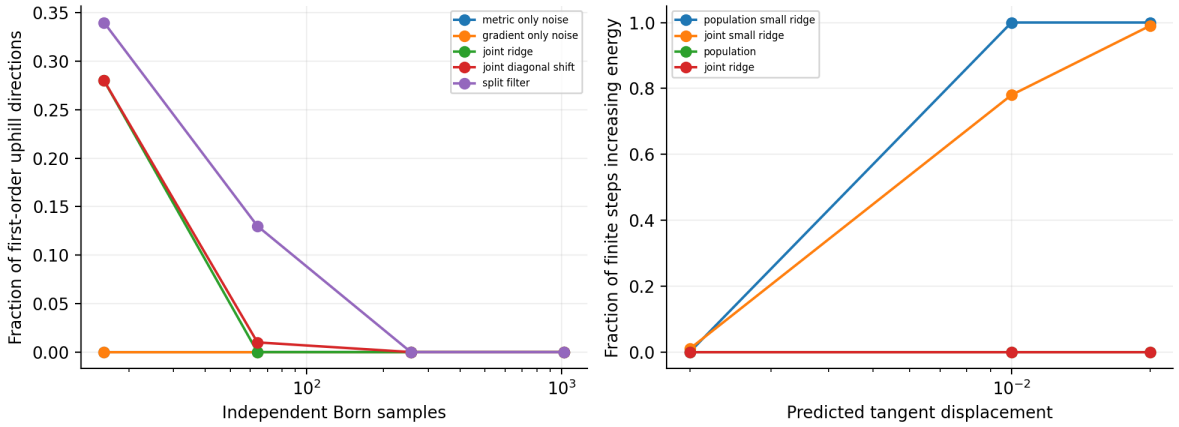


Figure 4: Left: first-order uphill fractions under independent metric/gradient interventions at checkpoint 25. Right: finite-step uphill fractions as the intended tangent displacement increases, with matched deterministic and sampled small-shift controls. Sampling error and nonlinear overshoot are separate mechanisms.

5.7 A descent direction is not a finite-step guarantee

Table 7. Weak-ridge linearization control at checkpoint 25. The ridge is $10^{-6}\bar{s}$. Sampled cases use 256 configurations. The actual distance is the mean phase-invariant chord distance; deterministic rows describe one fixed direction, repeated in the data structure for matching.

Intended distance	Update	Linear uphill (%)	Finite uphill (%)	Actual chord
0.002	population	0	0	0.004424
0.002	sampled	0	1	0.003426
0.01	population	0	100	0.1105
0.01	sampled	0	78	0.07107
0.02	population	0	100	0.4378
0.02	sampled	0	99	0.273

At intended displacement 0.02, the deterministic weak-ridge step changes the state by a chord distance about 0.438 and increases energy, despite being a true first-order descent direction. This is a failure of the local linearized step to predict the nonlinear displacement. It exists without sampling and therefore cannot be counted as evidence of a Monte Carlo failure.

Reducing the intended displacement to 0.002 eliminates deterministic overshoot for that control. The sampled weak-ridge update still has occasional finite-step failures, while larger-shift variants are more stable in this example. The result supports inspecting actual changes in the state and using finite-step safeguards, rather than attributing every unstable SR update to noisy eigenvalues.

5.8 Coordinate dependence of numerical rank and regularization

Table 8. Positive diagonal parameter rescaling at a fixed state. The physical family and formal tangent dimension are unchanged. Direction cosines compare each update with its own unscaled counterpart.

Scale extent (decades)	Raw rank	Normalized rank	Isotropic cosine	Diagonal-shift cosine
0.0	80	80	1	1
1.0	79	80	0.942	1
3.0	45	80	0.4464	1
6.0	24	80	0.3018	1

The raw thresholded rank falls from 80 to 24, while the column-normalized rank remains 80. This is not a loss of physical directions: the coordinate transformation is invertible. Isotropic trace-scaled ridge also changes the physical update substantially, because adding an identity matrix introduces a parameter-coordinate convention.

The diagonal-shift update agrees with its unscaled physical parameter step to relative differences below approximately 2×10^{-13} in the tested rescalings. That agreement follows from its covariance under positive diagonal transformations, and is supported numerically by a preconditioned solve. It does not make the method invariant under arbitrary parameter mixing, nonlinear changes of variables, or sample noise. A small isotropic numerical floor can also break exact covariance; it is not used in this idealized identity test.

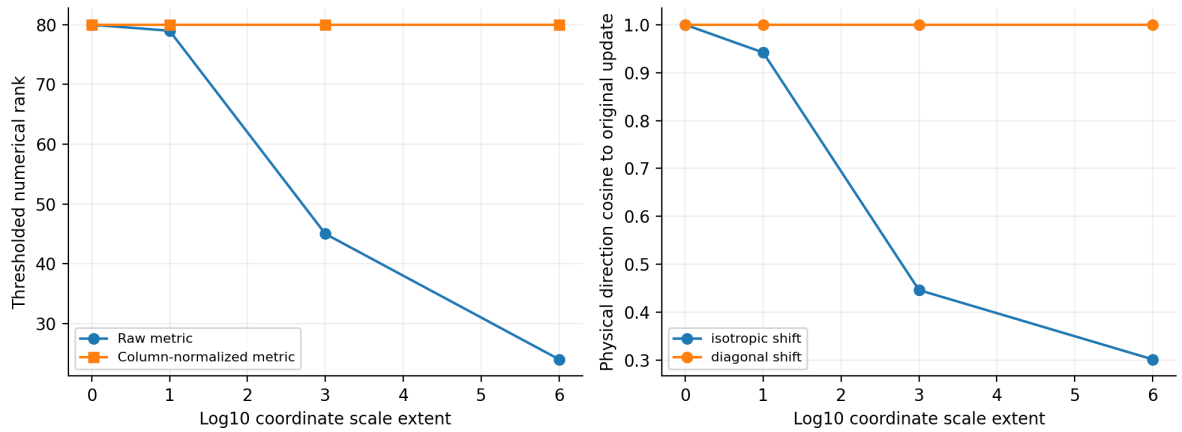


Figure 5: Thresholded rank and physical update direction under coordinate rescaling. The column-normalized metric removes this diagonal-scaling dependence, while the isotropic shifted update changes direction. These are numerical-coordinate controls, not changes in the represented quantum state.

5.9 An exploratory filter that is not consistently useful

The split-sample heuristic estimates candidate metric eigenvectors on half a batch, then retains directions only when their variance is within a factor of four on the second half and their projected gradient estimates have matching signs. It uses the same total sample count and population-scale ridge as the reference comparison. Appendix C gives the rule explicitly.

This heuristic is not a robust improvement. At checkpoint 25 and 64 samples, its first-order uphill rate is 13%, compared with 0% for the matched ordinary joint ridge. At larger sample counts it improves but still does not establish an advantage. Splitting a small batch sacrifices information, and directional sign agreement is too weak a guarantee of alignment with the population gradient. The current heuristic is therefore retained as a negative result rather than promoted as a new optimization method. Statistical filtering and independent-batch validation already have substantial prior context [6].

5.10 Exact-sum optimizer comparisons

Table 9. Exact-sum optimization from matched initializations, width eight. Each method allows 500 update steps. Energy and infidelity are seed means; Hamiltonian-vector counts include recorded line-search trials. Counts are not runtime or equal-cost guarantees.

Coupling	Optimizer	Energy error	Infidelity	H-vector calls
h=0.2	Adam	0.0001364	0.4974	500
h=0.2	SR isotropic	9.285e-07	0.004881	1000
h=0.2	SR diagonal	5.665e-07	0.004795	1000
h=0.2	SR block	7.737e-07	0.3819	1075
h=1	Adam	0.0008597	9.221e-05	500
h=1	SR isotropic	8.397e-05	8.068e-06	1000
h=1	SR diagonal	0.0001009	9.385e-06	1000
h=1	SR block	0.00011	1.141e-05	1035

SR improves the critical-chain energy substantially relative to the retained Adam settings. In the ordered chain, the full-metric SR variants also find much better target fidelities than Adam or the block approximation in these runs. Their energy differences are much less dramatic because of the small gap. These outcomes reinforce the need to report the target state or observable alongside energy.

The SR implementations use exact sums and an exact-energy Armijo backtracking test, with caps on predicted tangent displacement and parameter-step norm. Their stability cannot be credited solely to the metric, and the comparison is not a benchmark against tuned production implementations. The displayed Hamiltonian-vector counts omit different costs of Jacobian construction and metric solves. No end-to-end wall-clock speedup is inferred.

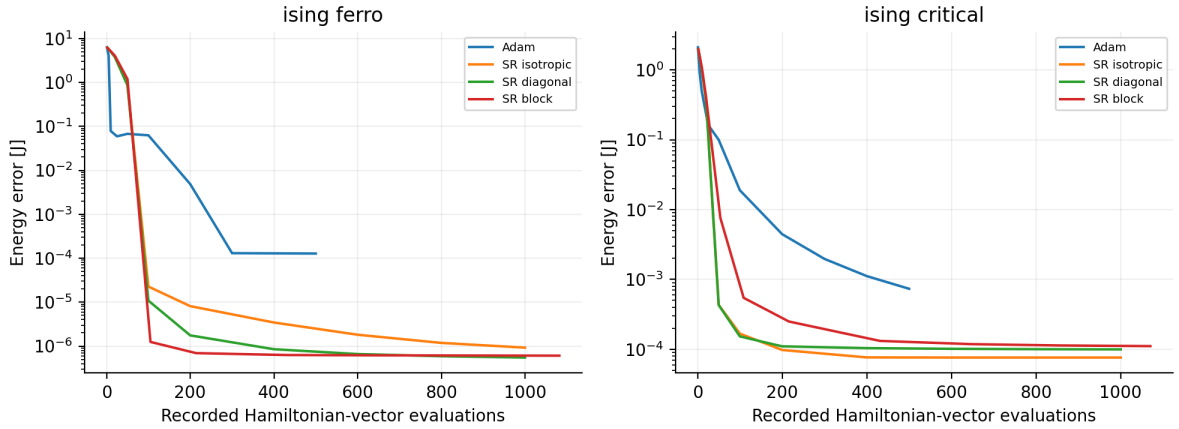


Figure 6: Representative seed-zero exact-sum optimization traces versus recorded Hamiltonian-vector evaluations. The comparisons use stated fixed hyperparameters and do not constitute equal-runtime studies. Both fields use an eight-hidden-unit RBM.

6. Discussion

6.1 A diagnosis should identify an intervention

The strongest statements in this study are constructive. A sign-mass bound certifies that the nonnegative family cannot approximate the selected frustrated target beyond a known fidelity. A lower-energy parameter vector in the same family certifies that a particular energy run has an optimization shortfall. Holding the exact wavefunction fixed while changing only the sampler isolates

mixing errors. Holding the checkpoint fixed while independently replacing metric and gradient estimates isolates sources of update error.

These tests do not yield a universal automatic classifier. Several mechanisms can coexist, and evidence about one does not exclude the others. A positive RBM can have both a sign obstruction and a poor optimization within its allowed orthant. A sampled update can suffer both gradient-estimation error and nonlinear overshoot. A low-gap objective can legitimately produce a useful low-energy state whose fidelity is unsatisfactory for a separate target-state task.

The practical sequence is therefore to ask what information would change the next action. A proven sign restriction calls for a different phase/sign representation or basis. A constructive lower-energy witness calls for an optimizer or initialization investigation. A frozen-state discrepancy between samplers calls for sampling changes. A gap-sensitive fidelity problem calls for a better specified state sector, observable, or objective. A raw rank statistic alone does not select among these actions.

6.2 What the QGT can and cannot diagnose

The QGT is an informative local geometric object. It quantifies state changes under parameter variations and underlies useful natural-gradient updates. Its singular directions can identify locally redundant parameter combinations, and its tangent image can reveal which part of an imaginary-time residual is currently accessible.

That local information must be interpreted with its statistical and coordinate context. A sampled metric can lose rank because only a few configurations were seen. A thresholded population spectrum can change under parameter rescaling. A full local tangent dimension can coexist with a global sign restriction. A large parameter update along a weakly visible tangent direction can invalidate the linearized geometry. None of these observations makes the QGT unhelpful; they specify the controls required to use it responsibly.

The practical recommendation is to report the normalized-state Jacobian or its equivalent metric convention, rank thresholds, sample and distinct-configuration counts, and both absolute and relative tangent residuals. When comparing regularizers, physical state changes and independent energy evaluations are more informative than parameter-step norms alone. These are reporting and diagnostic recommendations supported at small scale, not a new large-system optimizer.

6.3 Future directions

The results motivate three directions for further investigation.

First, symmetry identities can provide inexpensive sampler audits when energy is uninformative. If the trial family enforces a symmetry, observables odd under that symmetry have known zero expectation without exact diagonalization. Several such identities, paired chains, and sector-crossing counts could form a useful audit suite. The key qualification is that the identity must hold for the *current trial state*, not merely for the desired exact ground state. The frozen example demonstrates the principle; calibration during large-system training remains untested.

Second, a persistent bank of best-known parameter witnesses could make optimization diagnoses less dependent on a single training objective. In exact small systems, the energy comparison is direct. In sampled calculations, paired independent energy estimates and their uncertainty would be needed before declaring one parameter vector a better witness. Such a bank would bound optimizer shortfall from below without pretending to identify the global representation error. The present study provides a small exact example, not a general inference procedure.

Third, finite-step safeguards could test the actual state overlap rather than rely only on a linearized QGT norm. The weak-ridge control shows a large discrepancy between predicted and actual displacement even with exact statistics. An independent overlap estimate could therefore complement energy backtracking in a sampled optimizer. Whether it is reliable and affordable depends on its

own estimator variance and support coverage. No scalable overlap-based trust-region algorithm is demonstrated here.

6.4 Relationship to the earlier electronic-state study

Studies of geometry-conditioned electronic states emphasize that accurate scalar energy and a consistent local derivative construction do not automatically certify another observable. The present spin study identifies analogous distinctions for state fidelity, sampling, and tangent geometry. The implementations are separate: no molecular result is used as spin reference data, and no claim of transfer from electronic coefficients to the spin ansätze is made.

The methodological continuity is the use of exact small systems as a control, followed by interventions that expose the limits of a diagnostic. The next substantial extension should preserve those controls while introducing a larger configuration-conditioned model and actual sampled training. Simply scaling the current dense geometry diagnostics would not establish that the same conclusions dominate production variational Monte Carlo.

7. Limitations

The Hamiltonians are small one-dimensional spin models. Most systems have eight sites, with a ten-site state used for one training comparison and another for frozen sampling. There is no two-dimensional magnet, thermodynamic extrapolation, molecular simulation, or application to an experimentally unresolved phase. The sign obstruction concerns a real positive ansatz in a stated computational basis, not complex neural states in arbitrary bases.

The exact-sum training and supervised fits have finite budgets and only three initialization seeds. Many reach the cap, and successful numerical termination is not a global-optimality certificate. Supervised fitting uses exact labels and can be worse than energy fitting. The proven representation claim is limited to the sign-mass bound; all other achieved losses are finite-budget observations.

The sampling study freezes an exact ground state. Its energy estimator is deliberately an extreme zero-variance control. Actual imperfect states generally have nonconstant local energy, and local-energy diagnostics can then reveal some failures. The experiment proves that those diagnostics are insufficient in general, not that they are useless. Up/down trajectories use paired seeds, and no claim is made that a short global-flip repair handles more complicated multimodal distributions.

The stochastic metric study uses independent exact Born samples and only two checkpoints from one trajectory. It does not include MCMC autocorrelation in the metric/gradient batches. The shift scale and step normalization use population quantities available only in this diagnostic. Thus the one-step results are not deployment performance estimates or end-to-end sampled-training claims. The filter comparison has no extensive tuning or broader benchmark validation.

Coordinate invariance is tested only for positive diagonal rescaling. Column normalization and diagonal shifts do not solve arbitrary reparametrization problems. Numerical rank and tangent residuals require stated tolerances. The exact-sum optimizer comparison includes oracle energy line searches and different costs per update, so its iteration counts cannot establish a runtime advantage.

The literature already contains representation/sampling studies, QGT diagnostics, symmetry projection, and statistical SR regularization. The contribution of this study is a transparent collection of validated interventions and their measured outcomes. It does not claim a new representation theorem, a new universal failure taxonomy, or a state-of-the-art optimizer.

8. Conclusion

A stalled neural quantum-state calculation requires more than one diagnostic. In controlled small spin systems, we distinguish an analytically proven sign restriction, optimization shortfalls certified

by attainable parameter witnesses, mode-trapped sampling despite exact energy estimates, and objective sensitivity caused by a very small finite-size gap.

The QGT provides useful local geometry but does not by itself identify the mechanism. Its sampled rank can reflect observation count, its numerical spectrum can reflect parameter scaling, and its linearized norm can underestimate finite nonlinear state changes. Matched interventions separate noisy directions from deterministic overshoot, while the unsuccessful split-sample filter illustrates why a plausible diagnostic heuristic should be tested before being promoted.

The resulting research direction is concrete: combine physical-sector information, independent sampler audits, attainable-state witnesses, and measured state displacements. These controls offer a defensible foundation for larger sampled calculations, where the same questions become harder precisely because exact reference states are no longer available.

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Appendix A. Positivity bound and witness logic

Write $u = u_+ - u_-$ with disjoint nonnegative components. For any nonnegative normalized c, $u^T c = u_+^T c - u_-^T c$. Its positive value is at most $\|u_+\|$ and the magnitude of its negative value is at most $\|u_-\|$. Squaring gives $|u^T c|^2 \leq \max(P_+, P_-)$. The extremal nonnegative vector is the normalized larger sign component. Strict positivity allows the same supremum only as a limit if the other component is nonempty.

For any obtained witness θ_w, $\inf_\theta E(\theta) \leq E(\theta_w)$. Subtracting from $E(\theta_e)$ proves the optimization-shortfall bound. The witness need not be a ground-state approximation or a supervised optimum. If its energy is estimated stochastically, finite-sample uncertainty must be included before declaring a positive bound.

For the positive log-table family, differentiation gives $S_{ij} = p_i \delta_{ij} - p_i p_j$. The all-ones vector is null,

and $x^T S x = \mathrm{Var}_p(x) > 0$ for every nonconstant x when all $p_i > 0$. Hence its exact rank is $D - 1$, regardless of a global sign obstruction relative to a target outside the positive orthant.

Appendix B. Gradient and geometric conventions

All experimental states and parameters are real. The metric is the real normalized-state QGT; imaginary Berry-curvature components are not studied. For the RBM, the logarithmic derivatives are $O_{a_i} = \sigma_i$, $O_{b_j} = \tanh z_j$, and $O_{W_{ij}} = \sigma_i \tanh z_j$. Subtracting their Born means yields the normalized tangent. The energy-gradient convention contains a factor of two; SR matrices and gradients use that convention consistently.

The exact first-order energy change is $g^T \delta\theta$. The predicted tangent displacement is $\sqrt{\delta\theta^T S \delta\theta}$. The actual distance used in the tables is

$$d_{\mathrm{chord}}(c, c') = \sqrt{2(1 - |c^T c'|)},$$

which removes the unphysical global sign. It agrees with the tangent norm infinitesimally but need not do so for a finite parameter step. It is a chord distance between rays, not the exact Fubini–Study angle.

Under $\theta = D\phi$, $S_\phi = DSD$ for positive diagonal D. Adding $\lambda\,\mathrm{diag}(S_\phi)$ transforms in the same way as adding $\lambda\,\mathrm{diag}(S)$ before rescaling. An isotropic λI generally does not. Column-normalized solves implement the former identity without relying on numerically ill-conditioned raw coordinates.

Appendix C. Exploratory split filter and optimizer settings

The split filter divides each batch equally. It diagonalizes $\widehat{S}_A = U \Lambda U^T$ and evaluates $v_i = u_i^T \widehat{S}_B u_i$, $f_{Ai} = u_i^T \widehat{g}_A$, and $f_{Bi} = u_i^T \widehat{g}_B$. A direction is kept if

$$\lambda_i > 10^{-6} \bar{s}, \quad \lambda_i/4 < v_i < 4\lambda_i, \quad f_{Ai} f_{Bi} > 0.$$

The coefficient is $-(f_{Ai} + f_{Bi})/[2((\lambda_i + v_i)/2 + 10^{-3}\bar{s})]$, with zero assigned to excluded directions. The implementation uses no reference-gradient sign to select directions. The reference gradient is used only for scoring. Failure of this particular heuristic does not invalidate independent-batch validation more generally.

Adam uses learning rate 0.02, moments 0.9 and 0.999, and denominator offset 10^{-8}. Exact SR uses an initial multiplier at most 0.1, predicted tangent cap 0.05, parameter-step norm cap 1, and up to 25 Armijo halvings. Full and block isotropic shifts use $10^{-3}\,\mathrm{Tr}(S)/P$; the diagonal shift uses $10^{-2}\,\mathrm{diag}(S)$ plus a $10^{-10}\bar{s}I$ floor. These fixed settings are retained without a broad tuning claim.

Appendix D. Additional sampling and coordinate records

The step experiment comprises 28,800 scored records: two checkpoints, four sample counts, 100 batches, twelve update constructions, and three intended displacements. These are 800 independently seeded batches with paired methods and radii, not 28,800 independent statistical experiments. Population directions repeated in those records are deterministic controls.

Appendix E. Reproducibility

The implementation uses NumPy, SciPy, and Matplotlib. It does not use NetKet, JAX, PyTorch, or a high-level quantum-state framework. Generic L-BFGS-B and linear algebra are supplied by SciPy;

neural derivatives, Hamiltonians, Metropolis updates, Adam, SR, and all diagnostic interventions are implemented in the project code.

The numerical experiments use Python 3.12.14, NumPy 2.3.5, and SciPy 1.17.0, with one BLAS thread. Optimizer trajectories can vary across numerical libraries.

This manuscript package contains the LaTeX source and six figures; experimental code, saved parameters, raw numerical records, and run logs are not included.